

Boss of the SOC Version 1 (2015)

10 May 2026 23:30

Ransomware

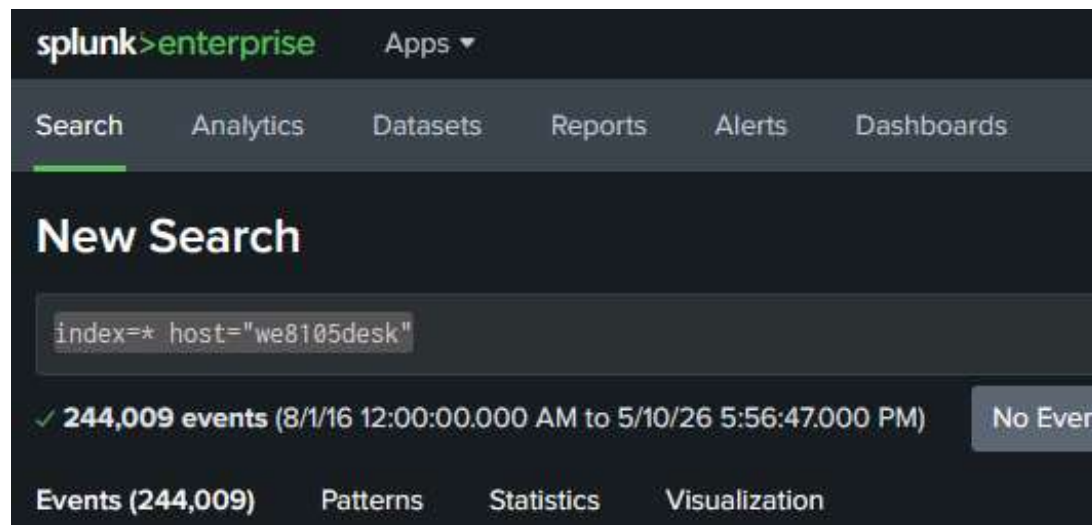
After the excitement of yesterday, Alice has started to settle into her new job. Sadly, she realizes her new colleagues may not be the crack cybersecurity team that she was led to believe before she joined. Looking through her incident ticketing queue she notices a "critical" ticket that was never addressed. Shaking her head, she begins to investigate. Apparently on August 24th Bob Smith (using a Windows 10 workstation named we8105desk) came back to his desk after working-out and found his speakers blaring (click below to listen), his desktop image changed (see below) and his files inaccessible.

Alice has seen this before... ransomware. After a quick conversation with Bob, Alice determines that Bob found a USB drive in the parking lot earlier in the day, plugged it into his desktop, and opened up a word document on the USB drive called "Miranda_Tate_unveiled.dotm". With a resigned sigh she begins to dig into the problem...

#200

Question:

What was the most likely IPv4 address of we8105desk on 24AUG2016?



Top 10 Values	Count	%
192.168.250.100	53,106	97.34%
192.168.2.50	1,217	2.231%
127.0.0.1	105	0.192%
192.168.250.255	70	0.128%
0.0.0.0	42	0.077%
224.0.0.252	6	0.011%
c0a8:fa64:0:0:9820:26fc:1e0:ffff	4	0.007%
::1	3	0.005%
fe80:0:0:0:9dac:222e:c1f8:d3d8	3	0.005%
239.255.255.250	1	0.002%

#201

Question:

Amongst the Suricata signatures that detected the Cerber malware, which one alerted the fewest number of times? Submit ONLY the signature ID value as the answer.

splunk>enterprise Apps ▾

Search Analytics Datasets Reports Alerts Dashboards

New Search

index=* sourcetype=suricata Cerber

✓ 5 events (8/1/16 12:00:00.000 AM to 5/10/26 6:08:46.000 PM) No Event Sampling ▾

Events (5) Patterns Statistics Visualization

```
> {"timestamp":"2016-08-24T10:49:24.542756-0600","flow_id":8
owed","gid":1,"signature_id":2816763,"rev":4,"signature":"
Show syntax highlighted
```

#202

Question:

What fully qualified domain name (FQDN) does the Cerber ransomware attempt to direct the user to at the end of its encryption phase?

index=* source="stream:dns" 192.168.250.100 "query_type()"~A cerber*

✓ 1 event (8/1/16 12:00:00.000 AM to 5/9/26 4:36:29.000 AM) No Event Sampling ▾

Events (1) Patterns Statistics Visualization

Format Timeline ▾ - Zoom Out + Zoom to Selection × Deselect

List ▾ Format 20 Per Page ▾

< Hide Fields All Fields

SELECTED FIELDS

- a host 1
- a source 1
- a sourcetype 1

INTERESTING FIELDS

- a app 1
- # bytes 1

i	Time	Event
>	8/24/16 5:15:12.668 PM	{ "endtime": "2016-08-24T17:15:12.668498Z", "timestamp": "2016-08-24T17:15:12.668494Z", "message": "Cerber ransomware encryption phase complete. Please visit the following website to retrieve your files: cerberhhyed5frqa.xmfir0.win", "query": ["cerberhhyed5frqa.xmfir0.win", "cerberhhyed5frqa.xmfir0.win"], "query_type": ["A", "Y"], "bytes": 90, "src_ip": "192.168.250.100", "src_mac": "0C:4D:E9:AA:B1:27", "src_port": 49456, "dest_ip": "192.168.250.20", "dest_mac": "00:0C:29:C3:C4:00", "dest_port": 53, "bytes_out": 0, "time_taken": 0.001 } Show syntax highlighted

host = splunk-02 source = stream:dns sourcetype = stream:dns

#203

Question:

What was the first suspicious domain visited by we8105desk on 24AUG2016?

New Search

```
index=* source="stream:dns" "query_type()"=A "192.168.250.100"
NOT query()="*.microsoft.com"
NOT query()="*.msn.com"
NOT query()="*.windows.com"
NOT query()="*.office.com"
NOT query()="*.live.com"
NOT query()="*.outlook.com"
NOT query()="*.bing.com"
NOT query()="*.skype.com"
NOT query()="*.zoom.com"
| table _time, query(), query_type(), dest_ip
| sort _time
```

✓ 33 events (8/23/16 12:00:00.000 PM to 8/25/16 12:00:00.000 AM)

No Event Sampling ▼

#204

Question:

During the initial Cerber infection a VB script is run. The entire script from this execution, pre-pended by the name of the launching .exe, can be found in a field in Splunk. What is the length of the value of this field?

```
sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" *.vbs app="C:\\Windows\\SysWOW64\\cmd.exe"
| eval lencmd=len(CommandLine)
| table lencmd
```

✓ 2 events (before 5/9/26 5:19:31.000 AM)

No Event Sampling ▼

#205

Question:

What is the name of the USB key inserted by Bob Smith?

New Search

```
index=* sourcetype="winregistry" host=we8105desk *usb*
| table registry_value_data
| dedup registry_value_data
```

✓ 519 events (before 5/9/26 5:27:12.000 AM)

No Event Sampling ▼

#206

Question:

Bob Smith's workstation (we8105desk) was connected to a file server during the ransomware outbreak. What is the IPv4 address of the file server?

New Search

```
index=* host=we8105desk
| stats count by dest_ip | sort - count
```

✓ 244,009 events (8/1/16 12:00:00.000 AM to 5/10/26 3:51:06.000 AM)

No Event Sampling ▼

#207

Question:

How many distinct PDFs did the ransomware encrypt on the remote file server?

New Search

```
index=* host=we9041srv  
| where like (Relative_Target_Name, "%.pdf")  
| stats dc(Relative_Target_Name)
```

✓ 525 events (8/1/16 12:00:00.000 AM to 5/10/26 4:06:41.000 AM)

No Event Sampling ▼

Events Patterns **Statistics (1)** Visualization

20 Per Page ▼

Format

Preview ▼

dc(Relative_Target_Name) ↕

257

#208

Question:

The VBScript found in question 204 launches 121214.tmp. What is the ParentProcessId of this initial launch?

New Search

```
index=* sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" vbs 121214.tmp
```

✓ 1 event (8/1/16 12:00:00.000 AM to 5/10/26 4:17:29.000 AM)

No Event Sampling ▼

Enable event sampling to run the search and return a random set of events.

Events (1) Patterns Statistics Visualization

Format Timeline ▼

Zoom Out

+ Zoom to Selection

× Deselect

#209

Question:

The Cerber ransomware encrypts files located in Bob Smith's Windows profile. How many .txt files does it encrypt?

New Search

```
index=* sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"  
TargetFilename="C:\\Users\\bob.smith.WAYNECORPINC\\*.txt"  
| stats dc(TargetFilename) AS distinct_txt_files_encrypted
```



The term 'TargetFilename="C:\\Users\\bob.smith.WAYNECORPINC*.txt"' contains a wildcard. Wildcards represent include punctuation. [Learn More](#)

✓ 406 events (8/1/16 12:00:00.000 AM to 5/10/26 4:47:27.000 AM)

No Event Sampling ▼

Events Patterns **Statistics (1)** Visualization

20 Per Page ▼

Format

Preview ▼

distinct_txt_files_encrypted ↕

406

#210

Question:

The malware downloads a file that contains the Cerber ransomware cryptor code. What is the name of that file?

New Search

```
index=* sourcetype=suricata dest_ip="192.168.250.100" file_state=CLOSED "http.hostname"="*.org"
```

✓ 1 event (8/1/16 12:00:00.000 AM to 5/10/26 5:51:36.000 AM) No Event Sampling ▼

Events (1) Patterns Statistics Visualization

Format Timeline ▼ — Zoom Out + Zoom to Selection × Deselect



#211

Question:

Now that you know the name of the ransomware's encryptor file, what obfuscation technique does it likely use?

Steganography